

DH2465 Pre-Study

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A structured pre-study covering thesis, solution hypothesis,
customer analysis, competitive landscape, business model,
risk assessment, and system design.

This document is a working draft intended to be iterated on. Figures marked with a source number are cited in the Sources section; unmarked figures are estimates or recommendations and should be validated before use in an investor or legal context.

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1. Thesis

Isolating the specific user friction point that this startup pre-study targets.

Health-conscious wearable users who suspect a connection between their daily schedule and their sleep or stress have no easy way to check why their performance scores change. Doing so requires manually comparing two separate systems, for example, a calendar and a health app, day by day over enough time to see a pattern. This is tedious and demanding enough that almost no one does it consistently, so suspicions become difficult to act on and follow up. Many users would also enjoy using their wearable for entertainment purposes, not only as a source of serious health data collection.

Wearables already generate scores and summaries, but only from the wearable's own sensor data. Nothing in that pipeline connects a health indicator to the reason behind it, because the reason usually exists in a completely separate system the wearable does not have access to. The wearables also do not produce any entertaining analysis regarding correlations between health indicators and other instances or events in the users daily schedule.

What would falsify this thesis (to test directly in discovery interviews):

- Target users report they've never actually wondered about these connections.
- Target users report they've tried manual comparison and found it easy or unnecessary.
- Target users say they'd rather ask a doctor/coach than see a pattern themselves (a real issue exists but the demanded solution is different from ours).

2. Solution Hypothesis

How the proposed product solves this friction in a way competitors cannot easily replicate:

We connect a user's calendar and their existing wearable(s) automatically, and run statistical analysis across both to surface specific, dated patterns (e.g., "meetings after 7pm correlate with 20% lower sleep efficiency") in a daily or weekly report, removing the need for manual cross-referencing that causes this friction today. Because the analysis could work across any wearable brand and treats the calendar as an equal data source, it produces insights that single-brand wearable apps or manual methods would struggle to replicate. We can also choose to connect our analysis to other apps to further make our analysis and reports more entertaining and unique.

Why incumbents are structurally unlikely to close this gap themselves:

The two data sources that matter (health, schedule) are owned by different companies with no shared incentive to combine them. Apple, Google, and the wearable brands each have a business reason to keep users inside their own ecosystem, not to make a competitor's calendar or a competitor's sensor data equally as important to the user.

Solution hypothesis weaknesses:

This is a temporary structural gap, not a permanent opportunity for technical competitive advantage. A large platform (Google Health API integrating calendar, or Oura shipping its own calendar feature) could close it. The durable opportunity, if one exists, is accumulated personal history and switching cost, not the cross-referencing mechanism itself. If the product is well-made and entertaining enough, users can be retained.

Core assumptions to test:

1. Users trust a third party with both calendar and health data enough to connect both.

2. The patterns found feel entertaining, interesting, and relevant.
3. Users change behaviour after seeing a pattern (not only information and insights).
4. A meaningful share of the target market is not already fully satisfied by their device’s built-in AI services.

3. Target Audience / Customer Analysis

3.1 Market segments

Segment	Description	Primary need
Performance athletes / biohackers	Already track HRV, recovery, training load closely	Precision, relevant correlation
Professionals managing stress/burnout	Wearable owners who sense work is affecting them physically	Validation, actionable change
People with sleep issues	Frequently use a sleep tracker already, may already show signs of anxious over-tracking	Reassurance, see orthosomnia risk
Casual quantified-self users	Enjoy data for its own sake, lower stakes	Novelty, fun, low commitment

3.2 Market sizing

Grand View Research puts the global wearable technology market at roughly **\$92.9 billion in 2025**, rising to about **\$229.97 billion in 2033** at a compound annual growth rate near 12% [1]. Fortune Business Insights’ smartwatch-only estimate for 2025 is **\$38.5 billion**, with an estimation to grow to **\$142 billion by 2034** [2].

These numbers vary a lot by research firm, which is normal for market-sizing reports (each uses different segment definitions and methodology). For the same broad “wearable technology” category and similar years, other firms report: IMARC Group, **\$72.5B (2024)** [3]; Value Market Research, **\$101.5B (2025)** [4]; Coherent Market Insights, **\$188.7B (2026)** [5]. We can gather a rough range, or magnitude (roughly \$90–\$190B depending on methodology) of the wearable market sizing and its growth potential over the foreseeable future.

Unit-shipment data from CCS Insight is more directly useful for sizing addressable *user* base than revenue figures, where overall smartwatch shipments are forecast to **decline roughly 4% in 2026** as replacement demand slows and prices rise, while smart ring shipments (Oura’s category) are forecast to **grow about 53% to 6 million units in 2026**, continuing at a projected 32% CAGR through 2030, against smartwatches’ 3% CAGR [6]. This suggests the highest-intent, most health-data-literate segment of the addressable market (ring wearers) is currently the fastest-growing one, even though smartwatches remain the larger installed base by volume and will stay as such for a while.

Recommended TAM/SAM/SOM structure (needs interview data?):

- **TAM (Total Addressable market):** All wearable owners globally (smartwatch + smart ring + fitness band), on the order of several hundred million active devices.
- **SAM (Serviceable Addressable Market):** Wearable owners who also use a digital calendar consistently and are in geographies where you can legally operate (start with EU + US/UK, given GDPR and CCPA both being addressable from one compliance framework).
- **SOM (Servicable Obtainable Market):** Realistic capture in year 1–2, based on comparable app category conversion rates (health/fitness apps typically see low-single-digit-percent

conversion from install to paid subscription — verify against direct comparables during competitive analysis, Section 4).

A realistic start for this pre-study is using students, friends, and relatives as early adopters to gather data and opinions.

3.3 Possible discovery interview plan?

- 15–25 interviews per segment.
- Ask about current behavior first: *“Have you ever tried to figure out if something in your schedule was affecting your sleep? What did you do?”* Don’t start with the product idea.
- Ask what they currently pay for related things (Oura/WHOOP membership, therapy, coaching apps, other subscriptions) to calibrate willingness to pay.
- Count how many describe an actual manual workaround attempt, this could be the strongest evidence for the thesis.

4. Competitive and Landscape Analysis

4.1 Direct competitors (wearable-native AI or analysis features)

Competitor	What they offer	What they don’t do
Oura Advisor	AI chat over the user’s own ring data	No calendar or external context
WHOOP Coach	AI coaching over recovery/strain/sleep data	Same limitation, single-device context only
Garmin / Connect app insights	Body Battery, Training Readiness scores	Single-device only, no external schedule context
Bearable, Gyroscope	Manual symptom/factor logging alongside some device sync	Relies on manual daily logging, which has the same drop-off problem as a spreadsheet

4.2 Indirect competitors

- A knowledgeable friend, coach, or therapist.
- A personal spreadsheet or Notion tracker (the workaround the thesis is built on).
- Doing nothing.

4.3 The biggest competitive risk

If Oura, WHOOP, or Garmin ship their own calendar-correlation feature, a large share of the differentiation disappears, since they already own the wearable relationship. Our most defensible responses are (a) multi-brand support, which none of the single-brand players is business-motivated to build, (b) a different, more entertainment-focused approach than what competitors offer, and (c) accumulated personal history, which is a switching cost that grows the longer a user stays.

4.4 Platform access constraints

Checked directly against current developer terms. This section should be re-verified periodically, as these programs change.

Platform	Current access status	Implication
Garmin	The Connect Developer Program’s application form has been removed, with Garmin telling applicants new API requests are paused while the program is “redesigned and modernized,” with no reopening date given as of September 2026 [7][8]	Cannot plan a direct Garmin cloud integration for launch, fall back to Apple Health/Health Connect data Garmin’s own app writes there
Oura	Oura’s API/MCP Agreement (effective 8 June 2026) prohibits aggregators from supplying Oura data to AI systems entirely, direct developers may input data into an AI platform only with the end user’s consent. Training, embeddings, and persistent storage of the data are prohibited for everyone. A separate clause requires prior written consent before charging users for functionality tied to the API [9][10]	The most restrictive and most nuanced of the platforms, needs confirmation directly with Oura in writing whether your specific processing (an LLM narrating pre-computed statistics, not raw data) fits the permitted “direct developer with consent” path before building on it
WHOOP	Development access is capped at 10 members until WHOOP approves the app for production. WHOOP’s own documentation gives no guaranteed review time, and its developer community forum shows multiple teams waiting two to four-plus months with no decision [11][12][13][14]. Separately, WHOOP’s API Terms of Use are reported as silent on AI/model use altogether, no explicit permission or prohibition [10]	Plan for a waitlist/manual-approval bottleneck before scaling past pilot users. WHOOP’s lack of information on AI use is not the same as permission, confirm in writing rather than assuming
Apple Health	No cloud API; on-device only, via HealthKit inside your own iOS app, permission granted per data type	Likely requires a real native/hybrid iOS app, not a website
Google Health API / Fitbit	The legacy Fitbit Web API is scheduled to be retired in September 2026; Google’s replacement, the Google Health API, is live now, but existing Fitbit OAuth tokens do not carry over, so every user must reconnect and re-consent [15][16]	Build against the current Google Health API, not legacy Fitbit documentation. Budget for a user re-consent flow if targeting Fitbit users specifically

For context on how differently platforms treat AI use of their data: one developer-facing comparison of six providers’ terms found Strava’s policy the strictest (prohibiting ingestion into a model’s context window at all), Garmin’s the most permissive (AI processing and training allowed with disclosure and consent), and WHOOP’s and Apple’s the least specified (neither mentions AI use explicitly) [10].

Recommended starting point: Apple Health, starting with Apple Watch data. Of the five platforms above, Apple Health is the only one with no approval gate at all, meaning there is no application to submit, no review queue, and no risk that an existing integration gets frozen mid-build, unlike Garmin’s paused program or WHOOP’s multi-month approval backlog. This works

for a resource-constrained early team, as every other platform in the table adds a dependency on a third party’s review timeline before any users can be onboarded. Apple HealthKit access is granted directly by the user, in the app. Starting here also avoids the largest open legal question in this document, which is restrictions on feeding API data into an AI system. HealthKit data is read on-device under Apple’s existing consent model rather than pulled through a developer agreement with AI-specific restrictions attached. HealthKit has no cloud API, so it likely requires a native or hybrid iOS app rather than a website, and Apple Watch data arrives as fairly raw signals (heart rate, HRV, sleep stages), meaning any “readiness” or “recovery”-style metric has to be computed in-house rather than imported. For a first version built with limited engineering time, a dependency-free integration surface is the more conservative choice, and the one most likely to generate early test findings from users. Garmin, Oura, and WHOOP support can follow once the approval and legal questions in this table are individually resolved.

No single platform is fully reliable right now. The MVP should lean on device-agnostic, on-device data (HealthKit / Health Connect) first, and treat each brand’s cloud API as an optional, separately negotiated add-on.

4.5 Legal and regulatory bottleneck check

See Section 6 for the full risk register. Items to resolve before committing engineering time: GDPR Article 9 status of health data [17], the EU AI Act’s treatment of health-adjacent AI outputs, and the wellness-vs-medical-device boundary for how insights are worded.

5. Business / Financial Model Design

5.1 Business Model Canvas (fill in as validated)

Block	Current best answer	Confidence
Value proposition	Automated cross-referencing of calendar and health data into specific and entertaining insights	Hypothesis, pending user test
Customer segments	Health-conscious wearable owners who already suspect a schedule/health link, stressed workers, casual for-fun users	Partially validated by thesis reasoning
Channels	App Store/Play Store, content marketing around insights and entertainment, partnerships with productivity or coaching communities	Unvalidated
Customer relationships	Self-serve subscription, automated reports, light-touch support	Unvalidated
Revenue streams	See Section 5.2	Unvalidated
Cost structure	See Section 5.3	Estimate only
Key partners	Wearable platforms (as gatekeepers until formal agreements exist), calendar providers, LLM vendor	—
Key resources	Statistical analysis engine, brand trust around health data handling, accumulated per-user history	—

5.2 Revenue model options

- **Subscription** (most likely fit for a report-based product with ongoing engagement).

- **Freemium** — free weekly report, paid daily reports or deeper analysis.
- **B2B2C** — corporate wellness programs, which is also where Garmin’s own enterprise Health API integrations are already sold, suggesting existing budget and buyer availability in that channel.
- **Data/research angle** — lowest priority given the API terms in Section 4.4 explicitly restrict downstream use of platform data.

5.3 Unit economics, categories to model

- **Cost per user:** LLM inference per report generated, hosting, push notification service, any per-connection fees from aggregators (e.g., Terra, Vital) if used instead of direct API integration.
- **Expected ARPU:** benchmark against comparable health/fitness subscription pricing (typically \$5–\$15/month in this category), verify against direct competitor pricing pages during Section 4.1.
- **CAC (Customer Acquisition Cost):** benchmark against consumer health app category norms, app-store-driven organic growth will have a materially lower CAC than paid acquisition, and should be the initial channel given a pre-revenue budget.

5.4 Cash requirement to reach MVP

Estimate based on: months of runway needed to complete the initial user test (near-zero cost) → MVP build (largest cost driver: mobile engineering time, not the AI itself) → first 100 paying users.

5.5 Human capital gaps to plan for

- Mobile/native engineering (iOS HealthKit + Capacitor or equivalent).
- Backend/data engineering (statistics pipeline).
- A design resource for the “fun” layer (Section 7) given its higher sensitivity.
- Access to a clinical or sleep-health advisor, flagged in the original risk register as missing, and worth resolving early given the orthosomnia risk below.

6. Risk Assessment

6.1 Risk register

No.	Risk	Category	Likelihood	Impact	Mitigation
1	Oura’s AI-use and monetization restrictions block or restrict the core feature	Platform/partner	High	High	Get written confirmation from Oura before building, design the product to function fully on HealthKit/Health Connect data alone
2	Garmin developer program remains paused through launch	Platform/partner	High	Medium	Do not plan a direct Garmin cloud integration for v1, use on-device Health data instead
3	WHOOOP approval delay blocks scaling past 10 pilot users	Platform/partner	Medium-High	Medium	Apply early, budget months of lead time, treat as a slow rollout rather than launch blocker
4	Apple/Google change HealthKit or Health Connect policy	Platform/partner	Low-Medium	High	Stay within documented, narrow data-type permissions; avoid dependence on any single undocumented behavior

No.	Risk	Category	Likelihood	Impact	Mitigation
5	Health data treated as GDPR special-category data (Article 9) without a proper legal basis	Legal/regulatory	High (near-certain applicability)	High	Explicit consent flow, DPIA before launch, data-processing agreement with LLM vendor
6	EU AI Act classifies health-adjacent AI output at a higher risk tier	Legal/regulatory	Medium	Medium-High	Legal review of specific feature set before launch, avoid medical diagnostic-sounding language by default
7	Insights read as medical claims, triggering medical-device regulation	Legal/regulatory	Medium	High	“Wellness pattern” positioning, explicit non-diagnostic language is important, legal review of all report copy
8	Insights feel gimmicky or repetitive after a few weeks	Market/adoption	Medium-High	Medium	Rotate insight types and track feedback ratings per insight, validate this in user tests before build
9	Users distrust AI-generated health claims	Market/adoption	Medium	Medium	Show confidence intervals/sample sizes, separate “fun” vs “serious” tiers by tone
10	Low willingness to pay	Market/adoption	Medium	High	Test real payment intent in initial testing phase, not just stated interest
11	Sparse/inconsistent data across devices	Technical/data quality	High	Medium	Require minimum sample sizes before showing an insight and handle missing data in UX
12	Statistical false positives from testing many pattern combinations	Technical/data quality	High if unaddressed	High (trust-destroying)	False-discovery-rate correction, confidence thresholds
13	LLM narration invents or distorts a number the statistics didn’t support	Technical/data quality	Medium	High	Validate every generated number against the underlying insight card before display
14	Runway exhausted before reaching sustainable revenue	Financial	Medium	High	Model Section 5.4 conservatively, keep the initial manual phase as long as possible before spending on engineering
15	LLM inference costs scale faster than revenue	Financial	Medium	Medium	Cap report frequency/length by tier, monitor per-user inference cost from day one
16	No clinical/sleep-health professional on the team or as an advisor	Team/execution	High (currently true)	Medium-High	Recruit an advisor before finalizing any health-claim wording
17	Ambiguous liability if an incorrect insight causes harm	Reputational/ethical	Low-Medium	High	Legal review of disclaimers and conservative wording. Red-flag detection routed to “see a professional” language, not a diagnosis
18	Friend-leaderboard feature processes a second person’s health-adjacent data without their direct consent	Legal/regulatory	Medium (if built)	High	Separate consent flow for any invited friend, do not inherit consent from the inviting user

No.	Risk	Category	Likelihood	Impact	Mitigation
19	Social comparison of sleep/stress metrics worsens some users' wellbeing	Ethical/reputational	Medium (if built)	Medium-High	Opt-in per metric, test with sleep-anxious users before shipping

6.2 On regulatory framing (baseline facts, not legal advice)

- Under GDPR, health data is treated as a “special category” of personal data under Article 9 [17], generally requiring explicit consent or another narrow legal basis, plus typically a Data Protection Impact Assessment before processing at scale. This is a foundational, stable requirement, not something that has changed recently, but the specific legal basis chosen should be confirmed with a lawyer once the exact data flows are fixed.
- The EU AI Act’s treatment of health-adjacent AI systems is an evolving area, confirm the current classification of your specific feature set with a lawyer close to launch, since guidance in this area continues to be clarified.
- None of the above replaces legal counsel, treat this section as the question list to bring to one.

7. The “Fun” Layer: Leaderboards, Playful Correlations, etc

This layer serves a **separate hypothesis** from the core thesis (retention/fun, not the schedule-health friction) and should be evaluated on its own terms.

7.1 Playful, non-medical correlations

Lower-risk, legally cleaner (explicitly non-causal framing), and gives users content on days with no “serious” finding. Main open question to test: whether a wrong “fun” correlation damages trust in serious insights sitting next to it in the same app.

7.2 Friend leaderboard

Highest-risk of the features:

- Processing a second person’s health-adjacent data for comparison purposes likely requires that person’s own direct consent under GDPR, consent cannot be derived from the person who invited them.
- Social comparison of sensitive metrics (sleep, stress) is a different risk category than comparing e.g. step counts, it can produce shame or anxiety rather than motivation, and no major sleep/recovery platform currently appears to offer this specific comparison. Worth confirming directly during competitive analysis, since its absence may be a signal.
- **Recommendation:** opt-in per metric, with sleep/stress comparison kept explicitly separate from lower-stakes metrics like step count or workout streaks.

7.3 Sleep companion mascot

Although an entertaining idea, this is the feature with the clearest documented risk (a mascot whose mood visibly worsens with a bad sleep score). **Orthosomnia** is a recognized phenomenon, first named in a 2017 case study by Baron et al. at Rush University Medical College and Northwestern’s Feinberg School of Medicine, published in the *Journal of Clinical Sleep Medicine* under the title “Orthosomnia: Are Some Patients Taking the Quantified Self Too Far?”, describing patients who became so focused on achieving a “correct” sleep score from their tracker that the resulting anxiety and perfectionism itself worsened their sleep [18]. A more recent peer-reviewed

cross-sectional study similarly reports notable rates of orthosomnia among sleep-tracker users, while its authors caution about the limits of that study design [19].

Design rules to adopt before building, not after:

- Default to encouragement-only mascot states, avoid visible “disappointment” tied to a bad score.
- Never let the mascot react negatively to nights plausibly outside the user’s control (illness, travel, a new baby) if that context is detectable from the data.
- Test specifically with users who already show signs of sleep-tracking anxiety, their reaction should decide whether this works at all.

7.4 Linking wearable health data with media consumption

Consumer wearables such as the Apple Watch continuously record physiological signals, including heart rate, heart rate variability (HRV), sleep stages, and activity levels. At the same time, streaming platforms like Spotify, Netflix, and YouTube log timestamped records of what a person listens to or watches. Because both data streams are time-indexed, they can be aligned on a shared timeline, which opens the door to exploring how media consumption and physiological state relate to each other in fun ways.

Health data can be exported from Apple Health as an XML archive or accessed programmatically through HealthKit. Media histories are available through official data-export requests (e.g., GDPR/CCPA data downloads for Netflix and Google Takeout for YouTube) or, in the case of Spotify, through its Web API and extended streaming history export. The resulting records are normalized to a common time zone and resampled to a shared resolution, for instance 1-minute or 5-minute windows.

Example analyses. Once aligned, a number of lighthearted but testable questions can be made:

- Does average heart rate differ between high-tempo and low-tempo music, using Spotify audio features such as *tempo*, *energy*, and *valence*?
- Is late-night streaming associated with shorter sleep duration or reduced deep-sleep fractions?
- Do suspenseful genres (e.g., thrillers or horror) coincide with elevated heart rate or reduced HRV compared to comedies?
- Do workouts accompanied by particular playlists show higher pace or sustained heart rate?

Methods. Simple approaches include windowed averages, lagged cross-correlation, and mixed-effects models that account for repeated measures within a person. For a fixed set of content categories, an event-study design (physiological signal before, during, and after a media event) offers an intuitive way to visualize effects.

Such correlations should, however, be interpreted with caution. Confounding factors (time of day, caffeine, physical activity, stress) are numerous, sample sizes from a single individual are small, and multiple comparisons inflate the risk of spurious findings. Moreover, causality cannot be inferred from observational data alone. A person may choose calming music *because* they are already relaxed. Finally, both health and viewing data are highly sensitive personal information, analyses should rely on data the individual owns, store it securely, and obtain explicit consent if data from other participants are included.

7.5 Falsification criteria for the fun layer as a whole

- If the fun layer measurably reduces trust in the serious insights, it needs tonal/structural separation (e.g., a separate tab or mode) rather than being scrapped outright.
- If leaderboard opt-in rates are very low or invited friends decline, the mechanic isn't wanted in this form.
- If mascot mood correlates with increased self-reported anxiety in testing, redesign is required regardless of engagement metrics.

8. System Design (conceptual level, precedes technical architecture)

- **Data collection principle:** prefer on-device extraction (HealthKit, Health Connect) over scraping or unofficial methods. Note that at least one open-source project resorts to browser automation to work around Garmin's lack of a self-serve API, this is a sign of how constrained that specific path is.
- **Database design:** structured, relational storage (e.g., PostgreSQL) is appropriate given the tabular, timestamped nature of health metrics and calendar-derived features.
- **Data provenance:** retain a source label on every metric from ingestion onward regardless of which platform terms end up applying. Several of the platform agreements reviewed in Section 4.4 (Oura and Garmin in particular) attach different obligations depending on how data is used downstream, so losing the source label early would make later compliance impossible to retrace. Re-verify each platform's exact attribution/branding requirements directly against their current developer terms before launch, since this document does not confirm the specific wording of every brand guideline.

9. Suggested Deliverables Checklist (status)

Deliverable	Status
Thesis	Drafted (Section 1)
Solution hypothesis	Drafted (Section 2)
Customer segments, personas, interview synthesis	Segments drafted; interviews not yet conducted
Competitive landscape matrix	Drafted (Section 4); needs direct pricing/review-data verification
Business Model Canvas + unit economics	Structure drafted; figures need vendor quotes
Risk register	Drafted with 20 entries (Section 6)
Regulatory question list for a lawyer	Drafted (Section 6.2)
Concierge/manual test results	Not yet run, highest-priority next step
Engagement hypothesis (fun layer)	Drafted (Section 7)
Go/no-go decision	Pending the above

10. Immediate Next Steps

1. Run the initial manual weekly-report test with 5–10 real users, using the interview questions in Section 3.3, before any further engineering commitment.
2. Send Oura a direct written question about whether your intended use (LLM-generated reports from their API data) is permitted under their current agreement. Do not build on an assumption here, given the specificity of the restriction found in Section 4.4.

3. Draft the GDPR consent and DPIA scoping document once your exact data flows are fixed enough to describe them precisely.
4. Recruit a sleep-health or clinical advisor before finalizing any report wording or the mascot's negative-state logic.

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Not independently sourced in this document (verify before relying on them)

- GDPR Article 9’s treatment of health data as special-category data is settled, stable law and is treated here as general legal background rather than a claim requiring a citation — confirm the specific application to your data flows with a lawyer.
- Pricing benchmarks in Section 5.3 and CAC/ARPU ranges are category-typical estimates, not sourced to a specific report, replace with real competitor pricing pulled directly from their sites during Section 4.1.